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A WEB-BASED SYSTEM FOR MANAGING STUDENT INDUSTRIAL ATTACHMENT RECORDS

BY

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Abstract

Industrial attachment is a mandatory component of undergraduate training that links academic learning to practical workplace experience. At present, the process of placing students, tracking their attendance and weekly logbooks, and evaluating their performance is handled manually using paper logbooks and spreadsheets, resulting in delays, lost records, and limited visibility for supervisors. This proposal outlines the development of a web-based Student Industrial Attachment Management System that will allow students to submit placement details and weekly reports online, allow institutional and industry-based supervisors to review and score submissions, and allow the department to generate consolidated attachment reports. The system will be developed using an Agile methodology, with a relational database back end and a browser-accessible front end. The expected outcome is a reduction in administrative workload and an improvement in the accuracy and timeliness of attachment records.

1 Chapter One: Introduction

1.1 1.1 Background of the Study

Industrial attachment (IA) exposes students to real work environments and is a graduation requirement in most Kenyan universities. Traditionally, students record daily activities in a physical logbook that is later reviewed by a university supervisor during a single site visit and marked upon return. This process depends heavily on paper records that can be damaged, lost, or delayed in transit, and it gives the department no way to monitor a student's progress during the attachment period. As student numbers have grown, the departmental attachment office now coordinates several hundred placements per academic year using spreadsheets and physical files, a workload that a centralized system could substantially reduce.

1.2 1.2 Problem Statement

The current manual system for managing industrial attachment records at the university is slow, error-prone, and difficult to monitor. Supervisors cannot verify a student's attendance or progress until the end of the attachment, placement records are scattered across departmental offices, and compiling attachment reports for accreditation takes several weeks of manual data entry. There is a need for a centralized system that allows real-time submission, supervision, and reporting of attachment records.

1.3 1.3 Objectives

Main Objective: To design and develop a web-based system for managing student industrial attachment records.

Specific Objectives:

1. To analyse the requirements of students, supervisors, and the attachment office for managing attachment records.
2. To design a system architecture and database for storing placement, logbook, and evaluation data.
3. To develop and integrate the web-based system based on the identified requirements.
4. To test and evaluate the system's functionality, usability, and reliability.

1.4 1.4 Research Questions

1. What challenges do students and supervisors face in the current manual attachment process?
2. What features are required to support online placement, logbook submission, and evaluation?
3. How effective is the developed system in improving record accuracy and reporting time compared to the manual process?

1.5 1.5 Significance of the Study

For students, the system will provide a single platform to register placements and submit logbooks without travel to campus. For supervisors, it will provide continuous visibility into student progress

rather than a single end-of-attachment review. For the department, it will reduce the time spent compiling attachment records for internal and accreditation reporting.

1.6 1.6 Scope and Limitations

The system will cover student registration for attachment, weekly logbook submission, supervisor evaluation, and report generation for a single university department. It will not process attachment allowances or integrate with external company payroll systems. Connectivity at some rural attachment sites may limit real-time logbook submission, which will be mitigated by allowing offline drafting before upload.

1.7 1.7 Definition of Key Terms

Industrial Attachment: A supervised period of practical work experience undertaken by a student in an organisation relevant to their field of study. **Logbook:** A record of daily or weekly activities undertaken during attachment.

2 Chapter Two: Literature Review

2.1 2.1 Overview of Attachment Management Practices

Several universities have moved from paper-based to electronic internship or attachment management to improve traceability and reduce administrative overhead (Mwangi & Kamau, 2022). Electronic logbook systems allow supervisors to review entries continuously rather than only at the end of the placement period, which studies associate with earlier identification of underperforming placements (Achieng, 2021).

2.2 2.2 Review of Related Systems

Generic learning-management and internship-tracking platforms exist, but are often either too broad for academic attachment requirements or too costly for institutional licensing. Institution-specific systems reviewed in the literature typically implement three core modules: placement registration, logbook/reporting, and supervisor evaluation, with role-based access for students, university supervisors, and industry-based supervisors (Otieno, 2020). Comparable systems developed for polytechnic internship programmes report reduced report-compilation time once logbook data is captured electronically rather than transcribed from paper (Njoroge, 2019).

2.3 2.3 Comparison of Existing Approaches

- **Paper logbooks:** Low cost and familiar, but not searchable, easily lost, and reviewed only at the end of attachment.
- **Spreadsheet tracking:** Improves aggregation for the attachment office but still relies on manual data entry and offers no student- or supervisor-facing interface.
- **General-purpose LMS internship modules:** Offer submission and grading workflows but are not tailored to placement-specific fields such as host organisation supervisor contacts or site verification.

2.4 2.4 Identified Gap

Existing literature confirms the value of digitising attachment management but provides limited detail on lightweight, department-specific implementations that integrate directly with a university's existing student records. This project addresses that gap by developing a system scoped specifically to the department's attachment workflow and reporting requirements.

3 Chapter Three: Methodology and Proposed Design

3.1 3.1 Development Methodology

The system will be developed using the **Agile (Scrum) methodology**, delivered in short iterations covering requirements analysis, design, implementation, and testing. Agile is appropriate because attachment-office requirements are expected to be refined through supervisor feedback during development, and iterative delivery allows core modules (placement registration, logbook submission) to be validated before evaluation features are built.

3.2 3.2 Requirements Analysis and Data Collection

Requirements will be gathered through structured interviews with attachment-office staff and a short questionnaire administered to students who have previously undertaken attachment, to identify functional needs (placement registration, logbook entry, evaluation) and non-functional needs (usability, data security, availability). Sample data-collection instruments are provided in Appendix A.

3.3 3.3 Proposed System Architecture

The system will follow a three-tier web architecture: a browser-based front end, a server-side application layer handling business logic, and a relational database storing student, placement, logbook, and evaluation records. Role-based access control will distinguish between student, university-supervisor, industry-supervisor, and administrator accounts, each restricted to the functions relevant to that role.

3.4 3.4 Tools and Technologies

- **Front end:** HTML5, CSS3, JavaScript (React)
- **Back end:** PHP (Laravel framework)
- **Database:** MySQL
- **Version control:** Git/GitHub

3.5 3.5 Testing and Evaluation

The system will be evaluated using unit testing of individual modules, integration testing across the placement–logbook–evaluation workflow, and user-acceptance testing with a sample of students and supervisors, assessed against functionality, usability, and reliability criteria.

3.6 3.6 Expected Outcomes

The project is expected to produce a working web-based attachment management system, a reduction in the time taken to compile departmental attachment reports, and documentation (user manual and system design report) to support handover and future maintenance.

4 Project Time Plan

Activity	Description	Duration
Proposal development	Title approval, background research, proposal writing and defence	Weeks 1–3
Requirements analysis	Interviews and questionnaires with attachment-office staff and students	Weeks 4–5
System design	Database design, architecture, and interface mock-ups	Weeks 6–7
Implementation (Sprint 1)	Placement registration and user-account modules	Weeks 8–9
Implementation (Sprint 2)	Logbook submission and supervisor-review modules	Weeks 10–11
Implementation (Sprint 3)	Evaluation scoring and report-generation modules	Weeks 12–13
Testing	Unit, integration, and user-acceptance testing	Weeks 14–15
Documentation & submission	User manual, final report compilation, and submission	Week 16

5 Estimated Budget

Item	Estimated Cost (KES)
Internet and data bundles	3,000
Domain and hosting (testing period)	2,500
Printing and binding	2,000
Travel for requirements interviews	1,500
Miscellaneous/contingency	1,000
Total	**10,000**

References

1. Achieng, P. (2021). *Electronic monitoring of student internships in Kenyan universities*. Journal of Higher Education Technology, 9(2), 45–53.
2. Mwangi, J., & Kamau, L. (2022). *Digitising work-integrated learning: A review of institutional practice*. East African Journal of Education Research, 5(1), 12–24.
3. Njoroge, D. (2019). *An electronic internship monitoring system for technical training institutions*. International Journal of Computing and ICT Research, 13(1), 33–41.
4. Otieno, R. (2020). *Design of an internship tracking system for higher education institutions* (Unpublished master's thesis). University of Nairobi.

Appendix A: Sample Data Collection Instrument (Extract)

Interview guide – Attachment Office Staff

1. How are attachment placements currently recorded and tracked?
2. What challenges do you face in monitoring students during attachment?
3. What information would you like a new system to report automatically?

Appendix B: Sample Weekly Logbook Entry Fields

- Student name and registration number
- Host organisation and department
- Week number and dates covered
- Summary of tasks performed

- Skills/tools applied during the week
- Industry supervisor’s remarks and signature field